

■ Learning Check

- (True or False.) According to the law of reflection, if the angle of incidence of a ray striking a mirrored surface is 35° , then the angle of reflection of the ray leaving the surface is 55° .
- The two types of reflection are _____ and _____.
- Two otherwise identical light waves arriving at the same point with peak matching valley undergo
 - constructive interference.
 - destructive interference.
 - polarization.
 - reflection.
 - None of the above.
- Light is totally absorbed when sent through two Polaroid filters if their respective transmission axes are _____.
 - perpendicular
 - parallel
 - at 45°
 - at 90°

ANSWERS: 1. False 2. specular, diffuse 3. (b) 4. perpendicular

■ Learning Check

- An object is placed 20 cm in front of a plane mirror. The image of the object formed by the mirror is located
 - 10 cm in front of the mirror.
 - 10 cm behind the mirror's surface.
 - at the mirror's surface.
 - 20 cm in front of the mirror.
 - 20 cm behind the mirror's surface.
- Parallel light rays from a distant object strike the surface of a concave mirror. After leaving the mirror's surface, the rays will converge toward the _____ of the mirror.
- (True or False.) A curved mirror that produces an enlarged image will also have a greater field of view than a plane mirror.
- The primary mirror in an astronomical telescope must be a
 - half-silvered mirror.
 - convex mirror.
 - concave mirror.
 - plane mirror.
 - hexagonal mirror.

ANSWERS: 1. (e) 2. focal point 3. False 4. (c)

■ Learning Check

- Upon entering a medium in which the speed of light is 1.5×10^8 m/s from a medium in which the speed of light is 2.0×10^8 m/s, an oblique ray will
 - be bent away from
 - be bent toward
 - travel along
 - travel perpendicular to
 - remain undeviated with respect to the normal to the interface between the two media.
- In air, a source of light has a wavelength of 590 nm. The same source under water will have (a) a longer wavelength, (b) a shorter wavelength, or (c) the same wavelength.
- The critical angle is that angle of incidence for which the angle of refraction equals _____.
 - 90°
 - 45°
 - 30°
 - 180°
- Name an important communication system component that makes use of total internal reflection.

ANSWERS: 1. (b) 2. (b) 3. 90° 4. optical fiber

■ Learning Check

1. (True or False.) A light ray can pass through a diverging lens without being deflected.
2. _____ lenses are thicker at their centers than at their edges, whereas _____ lenses are thinner at their centers than at their edges.
3. An image formed using a converging lens
 - (a) is always real.
 - (b) is always virtual.
 - (c) can always be projected on a screen.
 - (d) can have a negative magnification.
 - (e) can never be upright.
4. The magnification of an object by a certain lens is equal to -0.5 . The image formed by the lens must be
 - (a) upright and enlarged.
 - (b) upright and reduced.
 - (c) inverted and enlarged.
 - (d) inverted and reduced.
 - (e) virtual and unchanged in size.

ANSWERS: 1. True 2. Converging, diverging 3. (d) 4. (d)

■ Learning Check

1. A _____ lens is used to compensate for hyperopia.
2. When myopia is corrected surgically, the shape of the _____ is changed.
3. (True or False.) Blue light travels more slowly through glass than red light does.
4. A ray of violet light and a ray of yellow light enter a block of glass from air with the same angle of incidence. The angle of refraction for the violet ray will be
 - (a) much larger than that of the yellow ray.
 - (b) a little larger than that of the yellow ray.
 - (c) equal to that of the yellow ray.
 - (d) a little smaller than that of the yellow ray.
 - (e) much smaller than that of the yellow ray.

ANSWERS: 1. converging 2. cornea 3. True 4. (d)